CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

May 2014

EE405 Power System Protection

Date: 23.05.2014, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 [07]

i. Instantaneous over-current relays suffer from the problem of over reach because __________ present in the fault current.

ii. One disadvantage of using only time discrimination for transmission line protection is........ (Complete the sentence)

iii. The over-current relay is to be monitored by under-voltage relay when.......... (Complete the sentence)

iv. Buchholz relay is provided for __________ Protection.

v. Magnetising inrush current of transformers is rich in second harmonic contents. (state the sentence is true or false)

vi. Mho relay is used for short transmission line protection. (state the sentence is true or false)

vii. By burden of the relay, we generally mean VA rating of the relay. (state the sentence is true or false)

Q - 2(a) Figure 1 shows a single-line diagram of a portion of a power-system network. The relays shown are standard IDMT relays (setting range= 50 to 200% of CT secondary rating). The system permits continuous 10% overloading. The three phase short circuit current of bus A, B and C are 6000 Amp, 5600 Amp and 1900 Amp respectively. TMS of R_5 and R_4 are set as 0.1 each. The high set instantaneous setting of relay R_4 and R_5 are set at 800% and 2000% respectively. Determine the plug setting of all relays and time setting of R_1,R_2 and R_3. [08]

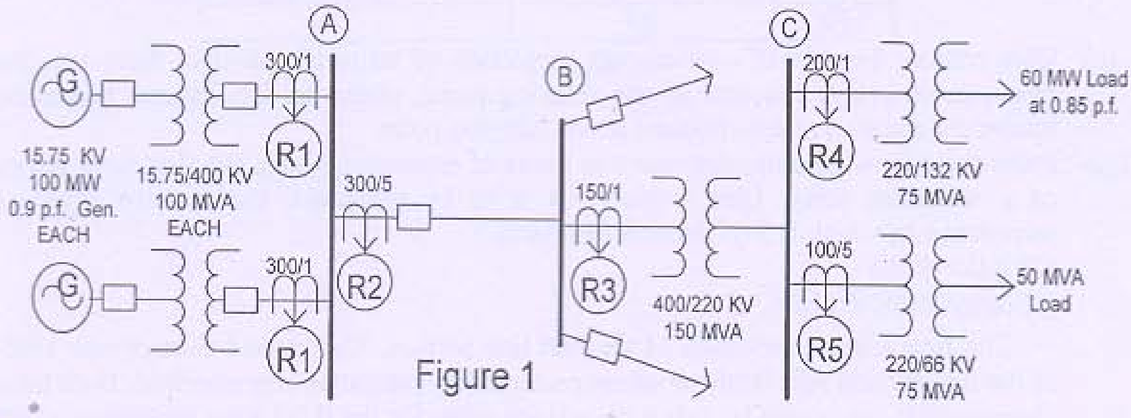

Figure 1

(b) Distance Relays cannot be applied for the transmission lines employing auto reclosers at both the ends of the line. [03]

(c) Give Reason: Over fluxing protection is provided for generator transformers only. [03]

OR

Q - 2(a) For the system shown in the Figure 2, all the relays are either directional or non-directional standard IDMT relays as per requirement. The fault level of bus A is 3500 MVA and that of bus B is 2500 MVA. Find the plug settings of the all Relays. Also find out the TMS of the all Relays if the TMS of R_2 and R_3 is set at 0.3 each. **[08]**

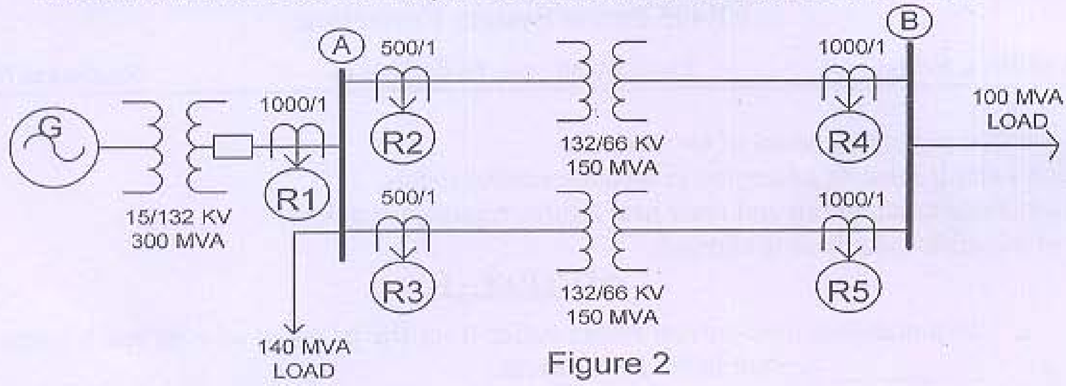

Figure 2

(b) For the power system shown in figure 3, complete the following table. **[03]**

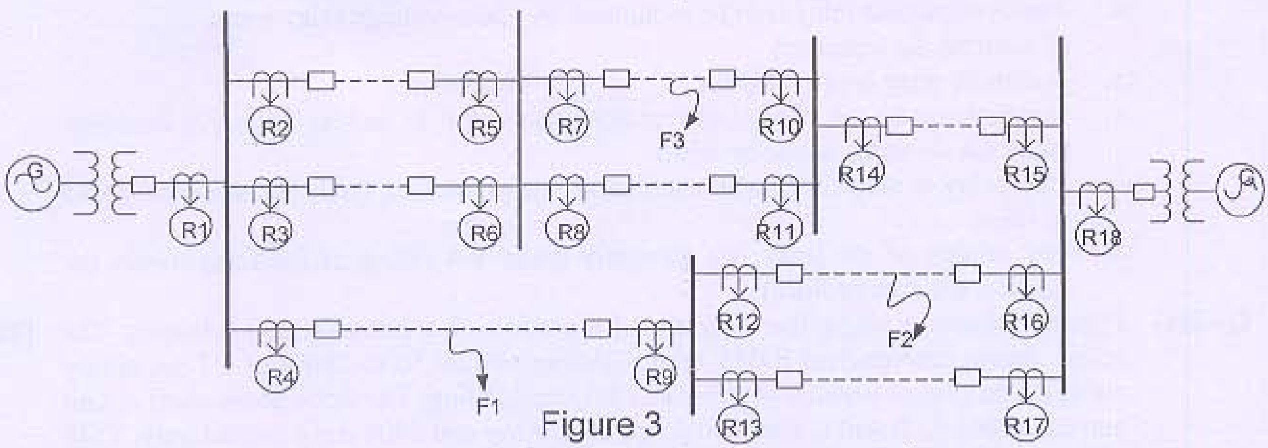

Figure 3

Fault at	Primary Relay fails	Backup relay/s operate
F1	R4	
F2	R16	
F3	R7	

(c) Give reason: In case of over-current protection of transmission line, there are four over-current relays required at one relaying point, where as for distance protection totally six distance relays required at one relaying point. **[03]**

Q - 3(a) From the following data, determine in terms of secondary ohms, the first zone setting of a reactance relay. Line section AB is to be protected, the positive sequence impedance of which is $3+j7.5$ ohms (primary). **[08]**

CT ratio=400/1

PT ratio=220KV/110V

The first zone covers 80% of the first line section. The second zone covers 150% of the line section AB. Both zones are protected by reactance characteristic. Both these characteristic are bound by using impedance relay for the third zone protection which covers up to 220% of AB. Find all three zone setting.

(b) Draw the block diagram of carrier current protection scheme and explain each block as well as component. **[06]**

OR

Q - 3(a) Explain in detail with suitable vector diagram: Inherent phase shift of currents in the transformer, which becomes obstacles in transformer differential protection. [08]

(b) Give reason: Instantaneous over current relays suffer from the problem of over reach. [03]

(c) Explain: For protection against phase faults, 90° connection is used, in case of directional relays. [03]

SECTION – II

Q - 4(a) Explain the reasons for using additional auxiliary contacts in Trip circuit of circuit breaker. [02]

(b) Explain the operation of Negative Phase sequence Relay with necessary diagrams. [05]

Q - 5(a) Explain the reason why we need to identify the fault even on the Neutral of Generator? Why portion of the winding remains un protected with a normal Earth fault protection? Explain the 100% Earth Fault Protection scheme to protect the Stator of Generator with necessary diagrams. [07]

(b) An 11 KV, three phase, 30 MVA, star connected alternator is protected by an earth fault relay having 10% setting. if the neutral resistance limits the maximum earth fault current to 40% of full load value, determine the value of the resistor and percentage of the winding protected. Find also the value of the earth resistor needed to allow only 9.5% of the winding to be left unprotected. CT ratio is 2000/1 Amp. [07]

OR

(b) With suitable diagrams explain the generator stator inter turn fault protection scheme. [07]

Q - 6(a) List out and explain in short: Abnormalities arises in induction motor. [06]

(b) List out and explain in short: Type tests that relay has to undergo. [08]

OR

Q - 6(a) Explain Frame Leakage Protection for busbar. [07]

(b) Explain step by step method to investigate problem present in Three Phase Over Current + E/F Protection Scheme with help of Primary injection method. [07]

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